1. **Group ID**: Group 3
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2. **Project Name**: OpenLLM Insight: A Visual Model Comparison System for LLM

3. **Target Data Set:** Hugging Face Open LLM Leaderboard
   The Hugging Face Open LLM Leaderboard is continuously updated as new models are evaluated. To ensure reproducibility and consistency of analysis, we will use a fixed snapshot of the leaderboard dataset taken at a specific date during the project timeline.

   Data will be accessed via the Hugging Face leaderboard's public dataset/API export and stored locally as a versioned CSV or JSON file.

   This snapshot approach ensures:

   - All visualizations are based on a stable dataset version.
   - Results are reproducible and comparable across users.
   - Future updates can be incorporated by explicitly refreshing to a new versioned snapshot.

   Short Description

   The Hugging Face Open LLM Leaderboard evaluates and ranks open-source large language models (LLMs) using standardized benchmark tasks. Each entry corresponds to a model and includes metadata such as architecture, parameter count, and license type, along with benchmark scores including MMLU, GSM8K, and TruthfulQA. The dataset enables structured comparison of model performance across reasoning, knowledge, and truthfulness benchmarks.

4.
   I. **Overview**

   This project proposes an interactive visualization system built on a fixed snapshot of the Hugging Face Open LLM Leaderboard to help practitioners explore, compare, and select open-source language models.

   The system will aggregate model metadata (architecture, size, license) and benchmark results (e.g., MMLU, GSM8K, TruthfulQA) into coordinated views that reveal trade-offs between performance, size, and usability.

   To quantify model quality, we will compute an overall performance score using a normalized benchmark aggregation method. Users will be able to filter by constraints such as parameter count and license, then visually inspect how candidate models perform across tasks.

The goal is to transform a dense leaderboard table into an exploratory tool that supports real model-selection decisions for research and production.

II. Methodological Clarifications

1. Definition of Overall Performance

To compute a single overall score per model:

Each benchmark score (e.g., MMLU, GSM8K, TruthfulQA) will be normalized to a 0–1 scale using min–max normalization across the snapshot dataset:

normalized_score = (score − min) / (max − min)

The overall performance score will be computed as:

- **Default:** Unweighted average of normalized benchmark scores.
- **Optional (Interactive Feature):** User-defined weights per benchmark to reflect task-specific priorities.

This approach ensures:

- Fair comparison across benchmarks with different scales.
- Transparency in how aggregate rankings are calculated.
- Flexibility for different use cases (e.g., reasoning-heavy vs. knowledge-heavy applications).

2. Operational Definition of Pareto Optimality

Definition:
A model is Pareto-optimal if no other model in the dataset achieves both:

- Higher overall performance, and
- Equal or smaller parameter count.

In other words, a Pareto-optimal model represents a trade-off frontier: improving performance would require increasing size, and reducing size would require sacrificing performance.

Operationalization in the Visualization:

We will plot models on a 2D scatter plot:

- X-axis: Parameter count

- Y-axis: Overall performance score

The Pareto frontier will be computed algorithmically by identifying models not dominated by any other model in both dimensions.

Frontier models will be visually highlighted (e.g., distinct color or boundary curve).

This allows users to quickly identify efficient models that maximize performance relative to compute size.

3 **Questions the visualization will answer**

- Which models achieve the best overall benchmark scores under specific constraints (e.g., ≤13B parameters, commercial license)?
- How does performance vary across benchmarks (MMLU, GSM8K, TruthfulQA, etc.) for a given family of models (e.g., LLaMA, Mistral)?
- What are the trade-offs between parameter count and performance, and which models lie on the Pareto frontier (i.e., are size-efficient)?
- For a given use case (reasoning vs. knowledge-heavy tasks), which models stand out as the best candidates under customizable benchmark weighting?

4. **Target users**

- ML engineers and data scientists choosing models for deployment or experimentation.
- Researchers comparing model families and scaling behaviors across benchmarks.
- Practitioners and students who want an intuitive overview of the open-source LLM landscape without manually parsing the raw leaderboard table.

5. **End goals of the project**

- Enable users to narrow down models satisfying constraints (compute budget, license, performance profile).
- Provide a transparent and reproducible aggregate performance metric.
- Highlight Pareto-optimal models to guide efficient deployment decisions.
- Deliver a reusable visualization tool that can be updated by importing a new versioned snapshot of the leaderboard.